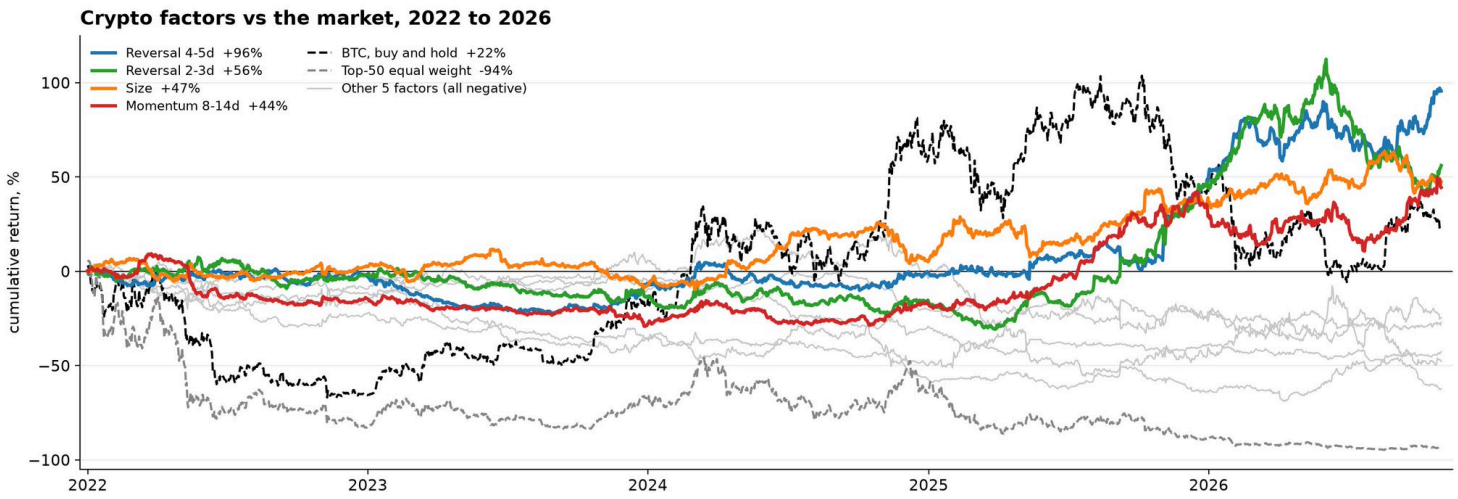
What drives crypto returns over time? A PIT Barra Factor Model

By Astraeus (@AstraeusCap). X article, 2 November 2026. Source: https://x.com/AstraeusCap/status/2117314412587483565

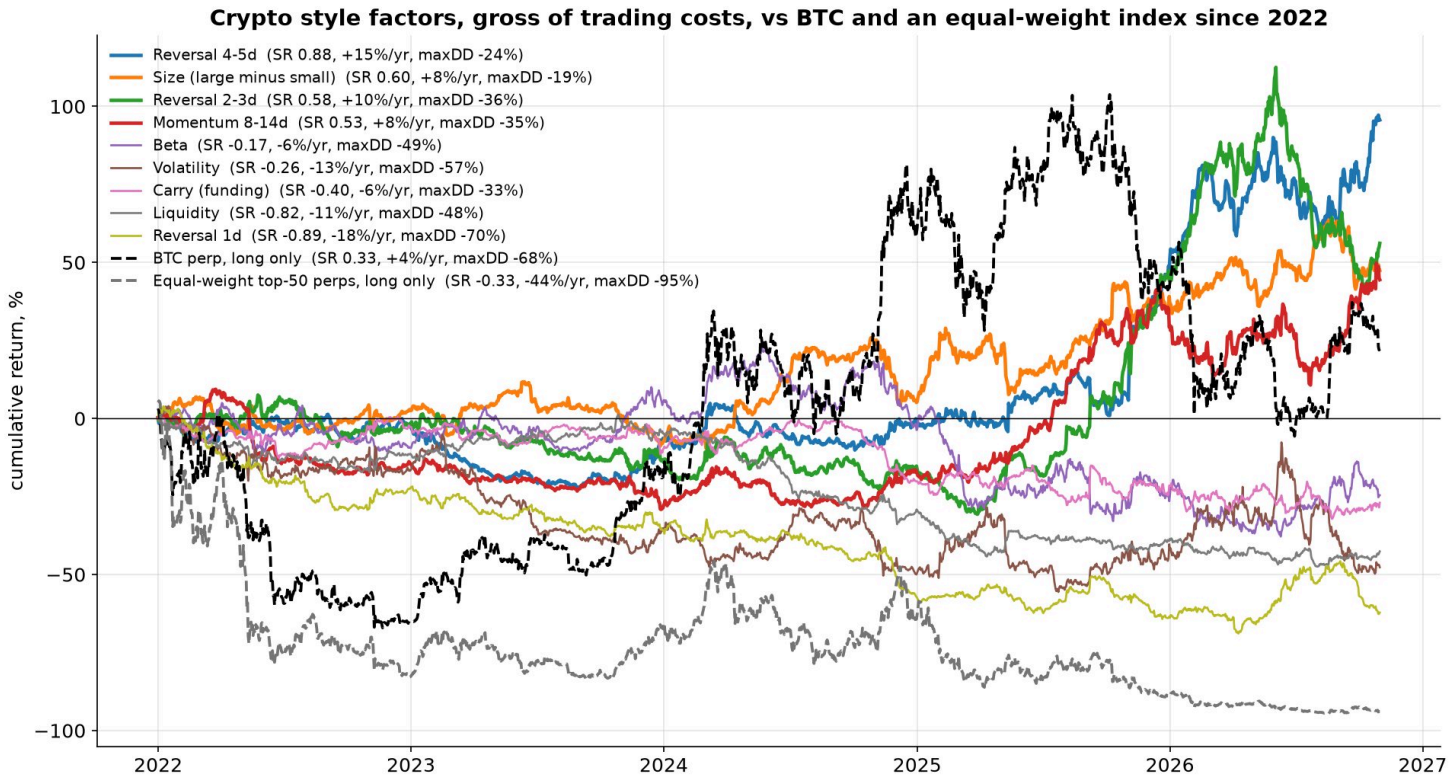
Archived copy rebuilt from the article text and its nine charts; chart placement follows the article captions.



Dollar-neutral factor portfolios, gross of trading costs, funding included. Binance + Bybit USDT perps, point-in-time top 100, out of sample 2022-01 to 2026-10.

Actual wannabe kwant stuff that might help you. Not indicator slop.

Since 2022, holding $BTC returned 22%, while an equal weight blend of the top 50 coins (rebalanced daily) returned -94%. A market neutral portfolio that buys coins that are recent losers and shorts ones that are recent winners, returned 94% with a third of bitcoin's drawdown (before costs). That's a factor, and while commonplace in TradFI, there isn't much discussion about them in crypto. I tried to find some common factors to see how much of returns they explain away.



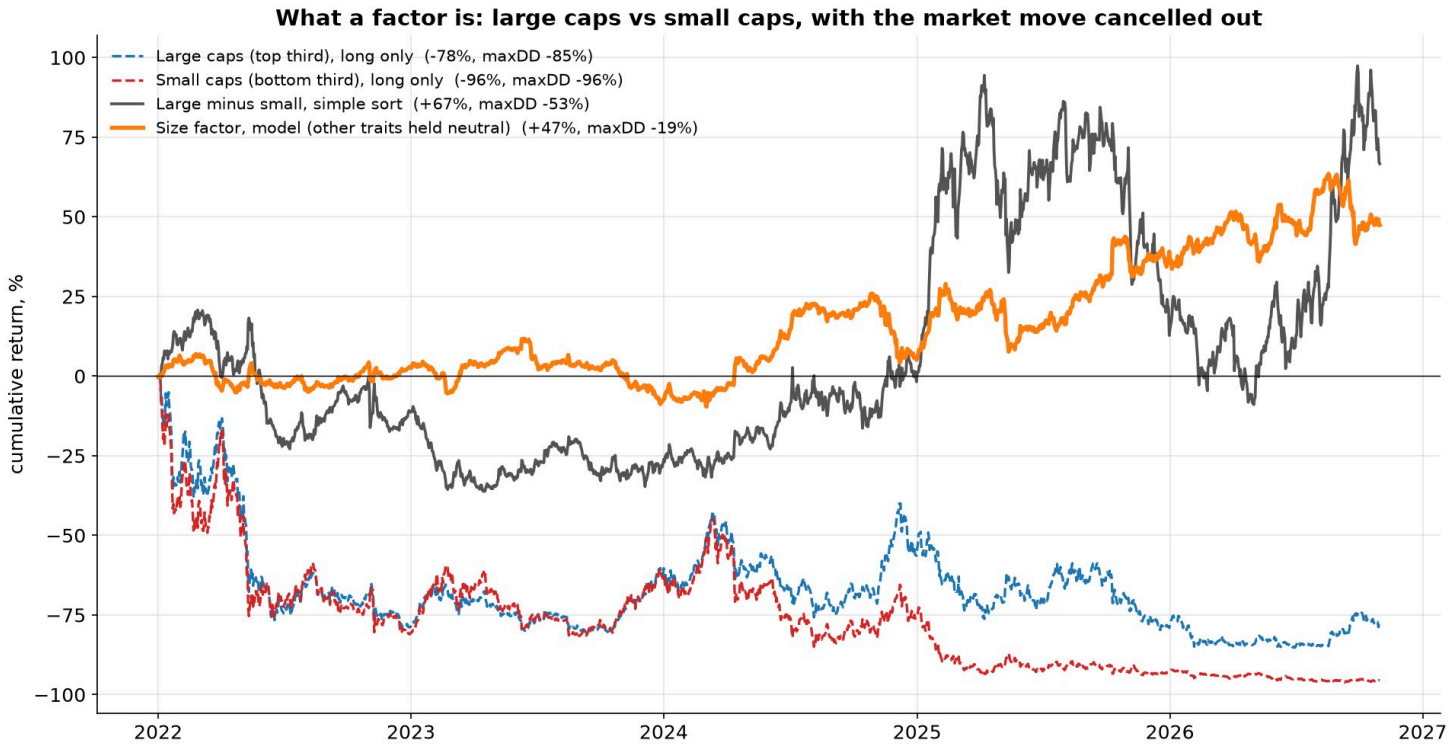
List of my factors and their return OOS 2022 to 2026, compared against $BTC 1x long and a top-50 equal weighted basket

I decided to try to start modeling factors as the first step in my personal research harness after reading threads from @therobotjames and @ScottPh77711570 who have a trove of old threads regarding basic effects, residuals, and how to think about modeling edge that helped me refine how I want to think.

What is a Factor?

A factor is just a common trait that many coins share and that can explain away some subset of returns. For example, in the image below you can see what happens when we sort every coin by daily market cap. Like I mentioned earlier, an equal weight basket of the top third rebalanced daily ends -78%, and the smallest third lost 96%.

Instead, if you long the top and short the bottom in equal dollar sizes (slightly simplified but that's the main idea), the market move cancels out and you can see the what's left of the size factor which is +67% from 2022.



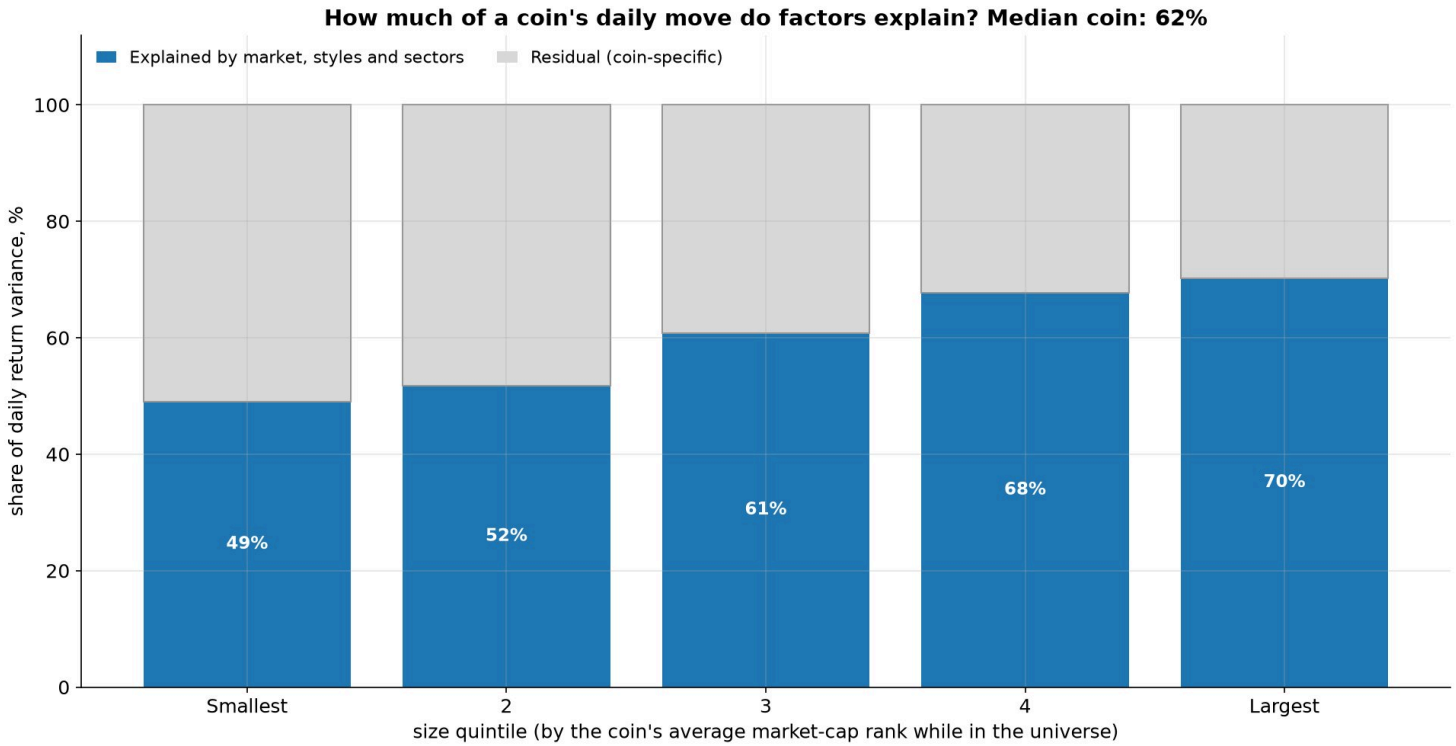
Large vs Small Caps Stripped Away

This isn't perfect however because this isn't normalized for other things that smaller caps may have in common, like volatility, sector, and other stupid stuff like how much they tend to clutter and are at the mercy of scam pump/dumps because usually they are worthless shitcoins (that somehow have a perp listing).

The easiest way to put it is like saying "tall people earn more". Tall people tend to be men, who tend to earn more, but if we did a simple sort on height we'd probably mix things up. If you regressed by height and gender you'd get a more accurate result. Following that same logic, the Barra model regresses the daily return of the coins in our universe on our factors, thus explaining how much each one matters. What it can't explain is the residual part, which is the coin specific part that you can find edge in.

Typically the sharpe of a factor doesn't matter, but knowing how many of us are just button clickers, I assume at least some of you will want to try to trade the factors themselves. That's why they're there but also as a way to try to explain how factors and drivers of returns change over time. For eg, beta barely moves but it does explain a lot about risk (MSCI's own crypto Barra model says this as well). Sharpe may just tell you what exposures pay you to trade them, how that may change, and lead to insights about deriving features from some of these effects that we can model and know exist.

How much is explained?



Share of each coin's daily return variance explained by the factor model by size quintile, 2022-2026

In our model, the median coin had 62% of their move explained by market, sector, and other associated factors. This splits between large caps (in which 70% is explained) and small caps (49%).

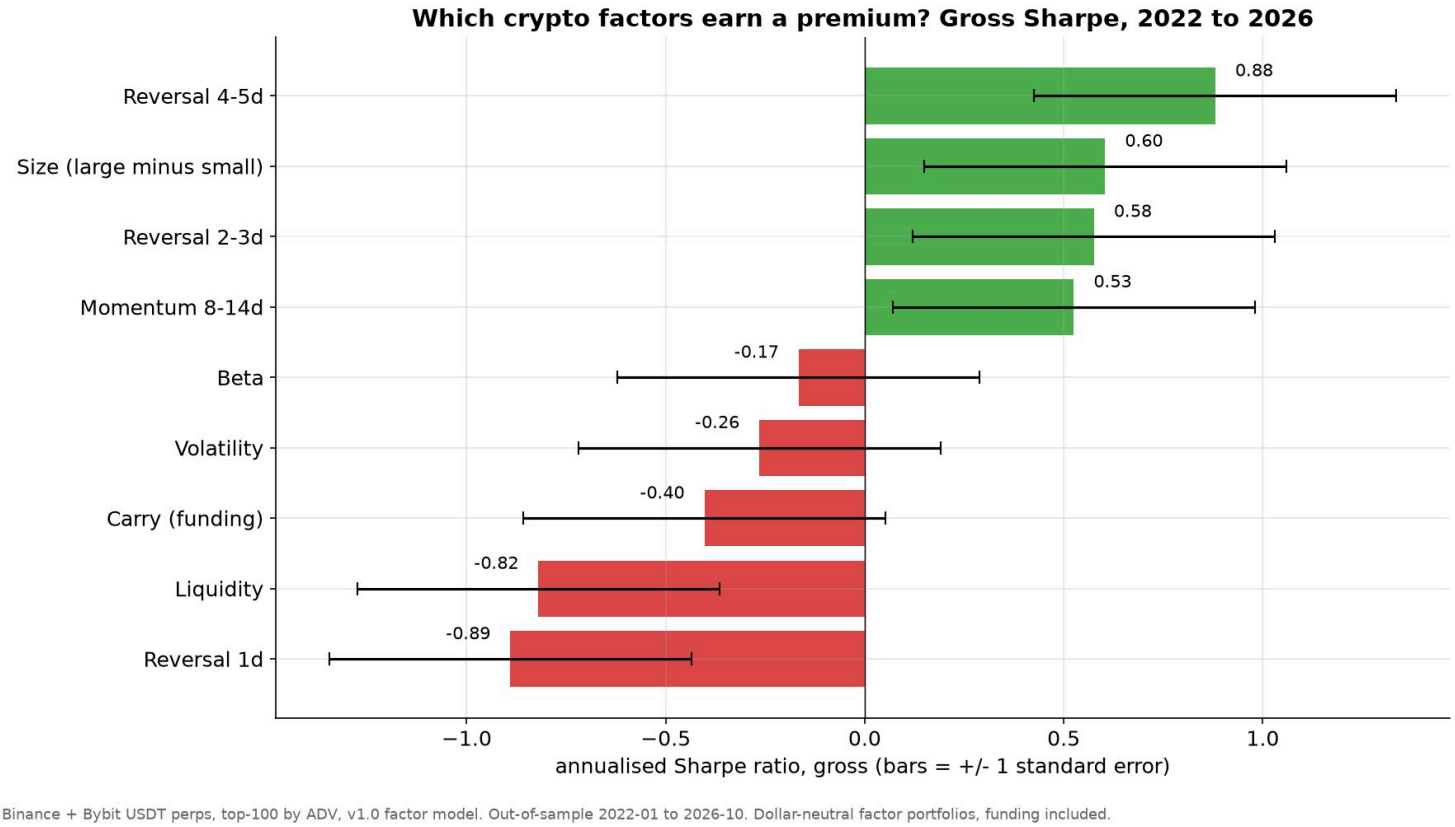
Nine Factors, but which ones paid?

The 9 factors we found were:

- Size (Large - Small market cap)
- Beta (Sensitivity to the market)
- Volatility
- Liquidity (Compared turnover)
- Carry (High - Low funding rate)
- Reversals (1/2-3/4-5d, recent losers - recent winners)
- Momentum (8-14d, winners - losers over two weeks)

This is by no means an exhaustive list, the market, sector, age since genesis block, and others are also likely other factors. Sector wide factors may also matter, i.e does an $ETH move explain away some EVM L2 moves? Updates to this will be coming shortly.

Before costs, three of these earned a premium: reversals, size, and momentum.



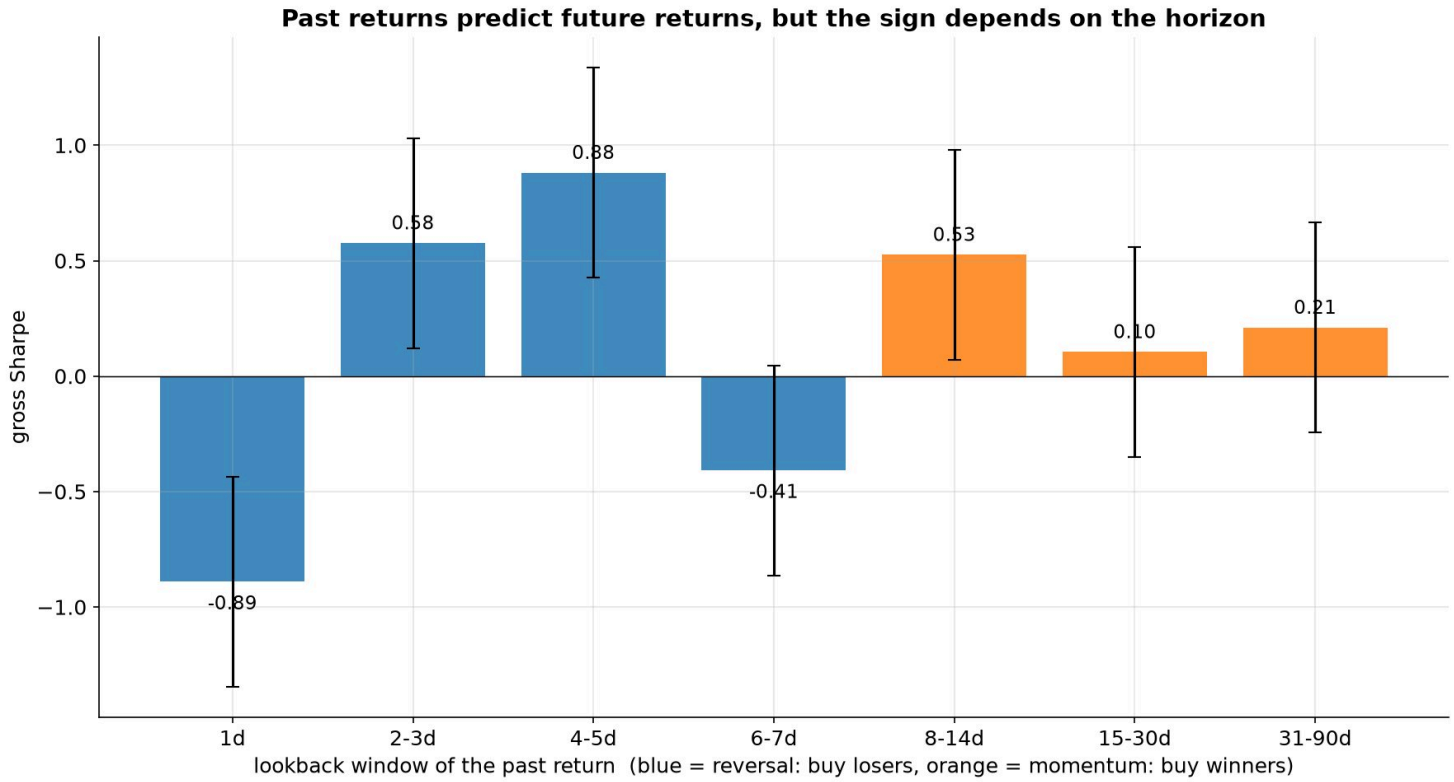
Out-of-sample Sharpe by factor, 2022-2026, before costs. Negative = the reverse bet paid

This is slightly misleading because the negative factors can be flipped into a premium. For example liquidity (-.82) just says that low turnover coins beat high turnover coins so the premium is in illiquidity. Reversal 1d (-.89) says that 1d momentum tends to continue. Similarly vol (-.26) and beta (-0.17) show possible low vol/low beta premia, but they're pretty small.

If you've made it this far, great! Now we get into the good stuff.

What lookbacks work?

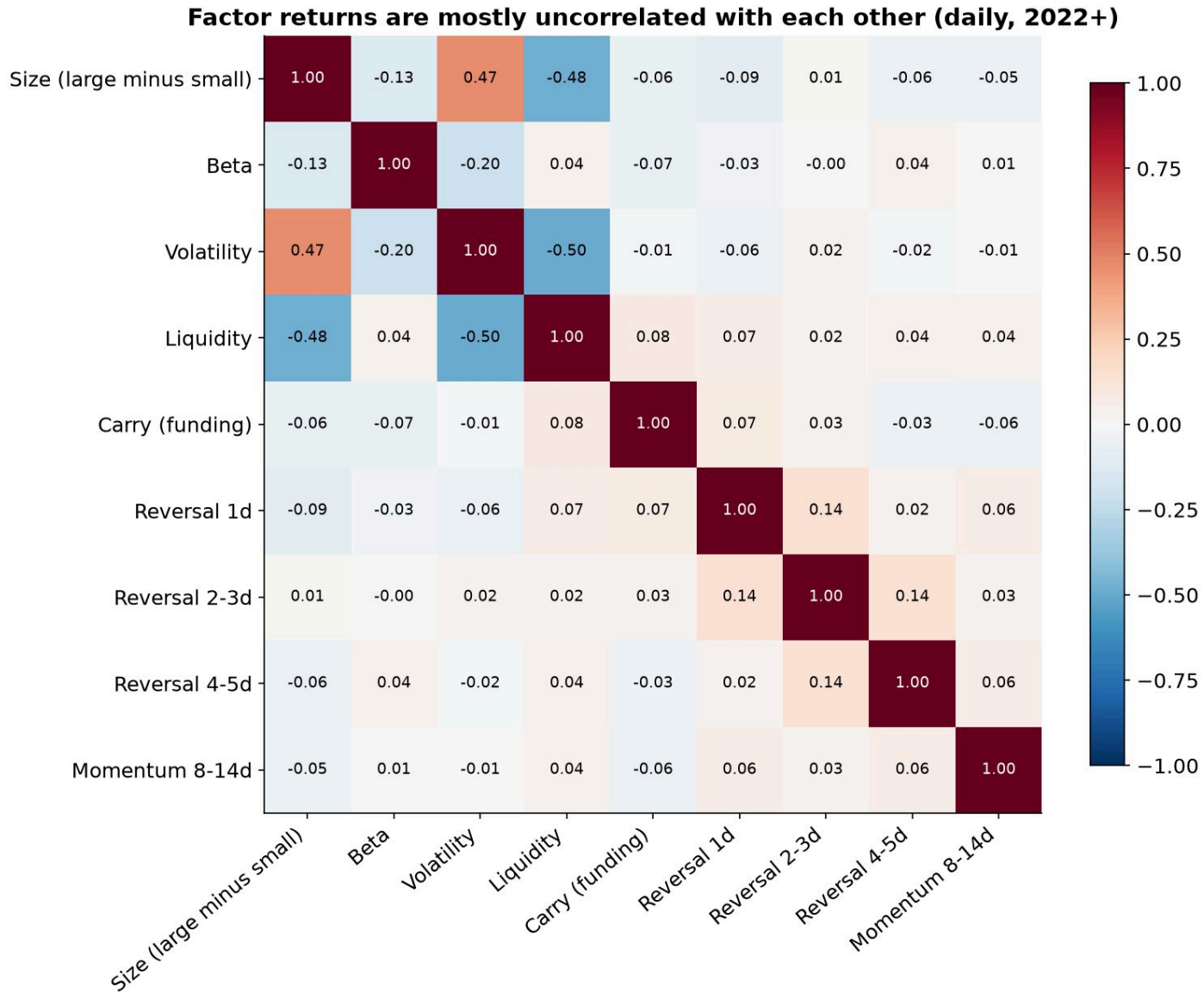
Past returns may have some predictive power over future returns, but the values are immediate and tend to stop over a multi day period. Winners tend to win for another day, but by days 3-5 they tend to mean revert. After 8-14 days, momentum starts to be more important.



Binance + Bybit USDT perps, top-100 by ADV, v1.0 factor model. Out-of-sample 2022-01 to 2026-10. Dollar-neutral factor portfolios, funding included.

Do factors move together?

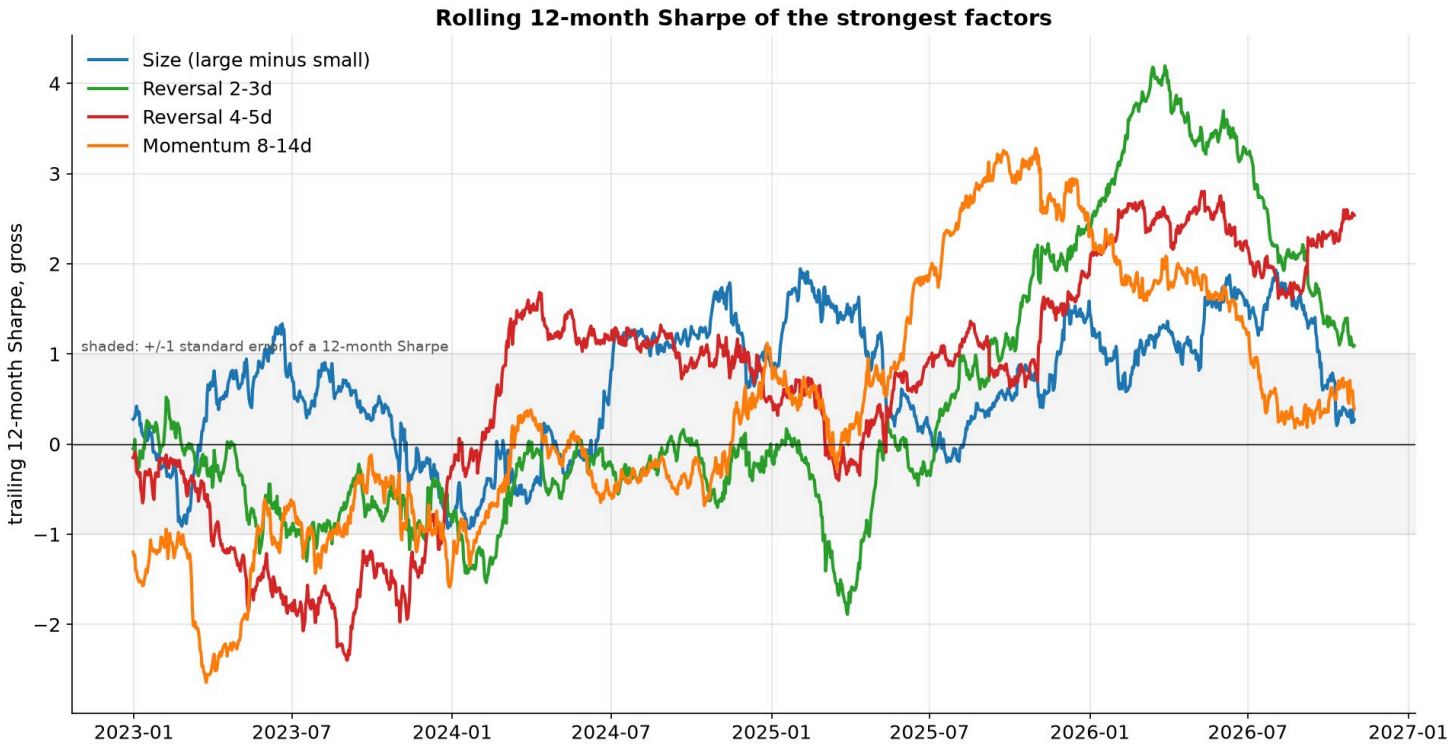
Our data says no, as evidenced by this factor correlation map. Median correlation between two factors are 0.06, with exceptions for size, volatility, and liquidity which again seems like a small cap overlap. In our case, low correlation is good because it means we're explaining away different parts of returns, and thus makes combining them more valuable.



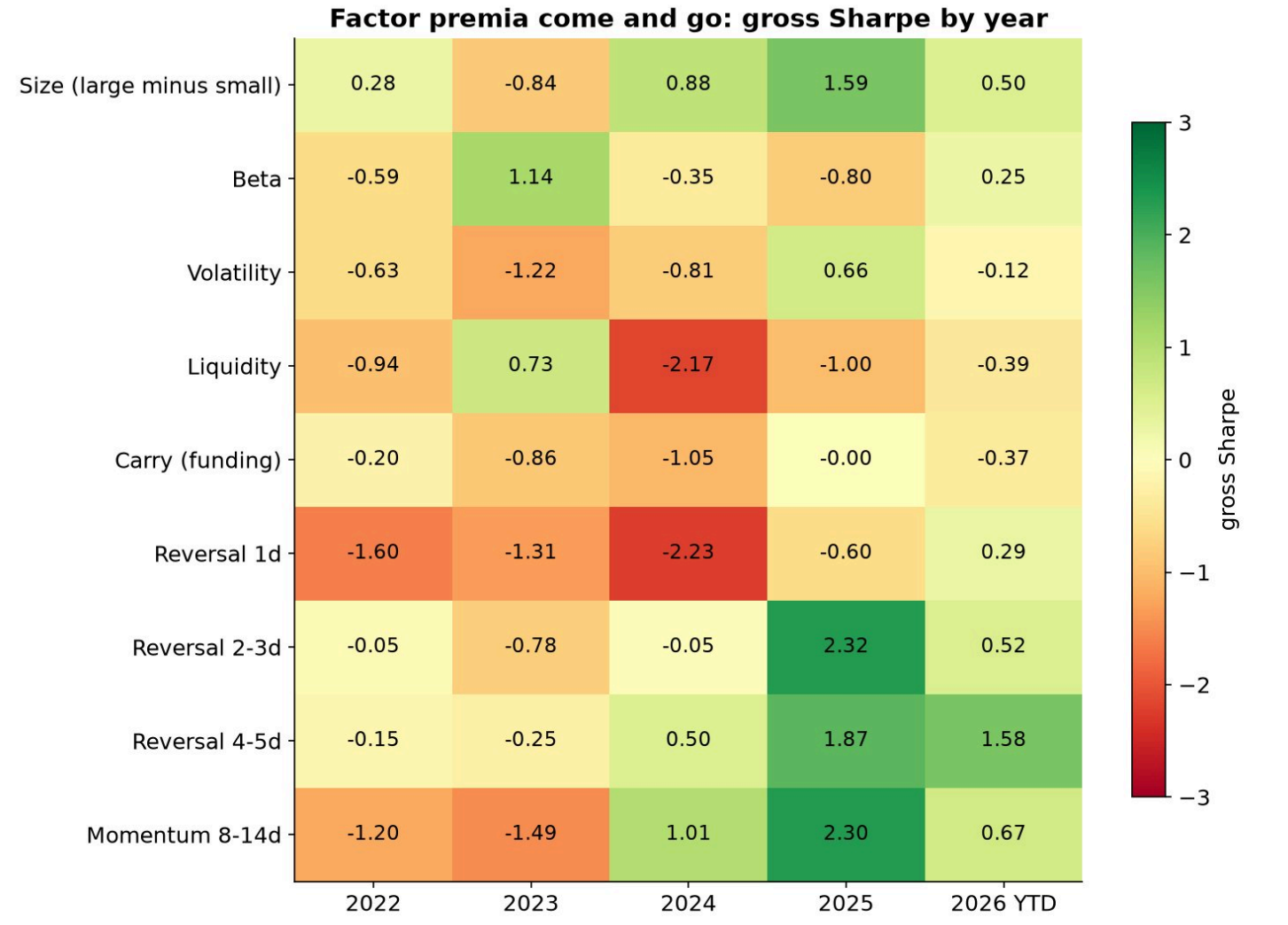
Binance + Bybit USDT perps, top-100 by ADV, v1.0 factor model. Out-of-sample 2022-01 to 2026-10. Dollar-neutral factor portfolios, funding included.

Is Premia Consistent y/y?

Premia is not very consistent year over year do the entrance of new market players, changes in crypto regimes, and changes over the broader global macroeconomic trend. Premia tends to come and go as you can see how reversal and momentum were flat/negative in 2022-23, but peaked on a 12 month rolling average in 2026 with a sharpe > 4, and yearly sharpes of 1.9-2. Size was also negative in 2013, but ended with a 1.59 sharpe in 2020.



Binance + Bybit USDT perps, top-100 by ADV, v1.0 factor model. Out-of-sample 2022-01 to 2026-10. Dollar-neutral factor portfolios, funding included.



Binance + Bybit USDT perps, top-100 by ADV, v1.0 factor model. Out-of-sample 2022-01 to 2026-10. Dollar-neutral factor portfolios, funding included.

The catch

On a rolling 12-month basis, a lot of these factors look like free money, but everything here is before trading costs. Like @ScottPh77711570 has said before, finding high sharpe isn't hard, but it's going to be a pain in the neck to trade. The next thread is going to be on why execution / turnover eats the edges (if you were to just try to trade factors), and how we can try to remedy that so we can derive signals from the factors that might be a bit better to trade.